

Linux Administration Practical Lab Worksheet

Topic: File System Navigation, File & Directory Management, User Management, and Permission Management

Duration: 2 Hours

Learning Outcome: Students will demonstrate competence in navigating the Linux file system, managing files and directories, handling users and groups, and controlling permissions.

Exercise 1: File System Navigation

Objective: Practice navigating and locating files using Linux commands.

Tasks:

1. Move to home directory and display current path.
2. List all files including hidden ones.
3. Navigate to /etc and list its contents.
4. Search for 'passwd' in the system.

Exercise 2: File and Directory Management

Objective: Create, copy, move, and delete files and directories.

Tasks:

1. Create directory structure ~/practice/linux/{configs,docs,logs}.
2. Create, copy, rename, and delete files as instructed.

Exercise 3: User Management

Objective: Manage user and group accounts.

Tasks:

1. Create user student1 and group classA.
2. Add user to group, verify membership, and delete user.

Exercise 4: Permission Management

Objective: Modify and verify file permissions.

Tasks:

1. Create project.txt and adjust permissions using chmod and chown.

Exercise 5: Combined Challenge Task

Objective: Apply all previous skills.

Scenario:

1. Create /home/admin/project_team/.
2. Add users alice, bob, carol and group team.
3. Set ownership and permissions on plan.txt file.

Instructor Evaluation

Criteria	Marks
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Correct command usage	5
Proper output/results	5
Understanding of concepts	5
Neatness & organization	5
Total	20 Marks